

Subword-Based Comparative Linguistics across 242 Languages Using Wikipedia Glottosets

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Abstract

We release MORPHOBPE, a framework for training and comparing per-language BPE tokenizers across 242 Latin- and Cyrillic-script languages built from Wikipedia *glottosets* (per-language TF-DF lexicons), with four diagnostic evaluations of what BPE captures across scripts. Unsupervised language identification by token count reaches 13.8% accuracy across 321 Latin-script languages, against 19.83% for supervised fastText lid.176, which returns zero for 147 of them because they fall outside its 176-language alphabet, including our best performers (Lak 81.5%, Cree 80.6%, Inuktitut 63.8%, Kabardian 60.1%). BPE aligns with morpheme boundaries at F1 = 0.28 across 15 languages (Morfessor: 0.43), and beats supervised character-*n*-gram cross-lingual transfer by +37% F1 across six pairs where labeled morphology is unavailable. BPE distance correlates moderately with Glottolog phylogeny (Mantel $r = 0.329$). We position per-language BPE as a coverage-first diagnostic for languages outside the supervised long tail, not as a substitute for supervised methods where labels exist.

1 Introduction

Traditional comparative linguistics, while providing deep historical and typological insights (see Lehmann 2013; Beekes 2011; Nichols 1992; Partanen et al. 2021; Säily et al. 2021), often lacks the scalability to handle the current volume of digital text (Arnett and Bergen 2024; Akindotuni 2025). However, recent NLP advances facilitate massive, data-driven studies (see Dang et al. 2024; Bender 2011; Imani et al. 2023; Sproat 2016), revealing universal tendencies in phonetics (Blum et al. 2024), lexical semantics (Tjuka et al. 2024), and sound symbolism (Ćwiek et al. 2022; Cathcart and Jäger 2024) that were previously inaccessible through manual, small-scale methods.

Despite this progress, large-scale studies frequently neglect endangered languages (Hämmäläinen

2021) and systematic performance disparities in multilingual models (see Shani et al. 2026; Chelombitko et al. 2024; Dunn et al. 2011). We address this gap by examining 242 languages¹ through a novel script-level lens (Latin vs. Cyrillic), moving beyond family-specific comparisons (Meillet 1967; Beekes 2011) to reveal overarching patterns visible only when languages share a script-based writing system (Daniels and Bright 1996; Rust et al. 2021).

In particular, we adopt a subword-based strategy (Sennrich et al. 2016) using Byte-Pair Encoding (BPE) (Gage 1994) tokenizers, which we train both on individual languages and on large aggregated corpora for each script. Aggregating data for all Latin and Cyrillic-script languages into respective training sets effectively unifies each script community into a single model, leveraging the shared-parameter paradigm common in massively multilingual pretraining (see Johnson et al. 2016; Conneau et al. 2020).

Although subword units are not perfect morphemes (see Sennrich et al. 2016; Bostrom and Durrett 2020), they serve as robust automated tools for macro-linguistic research (see Khurana et al. 2024; Pham et al. 2024; Futrell et al. 2015) that scale more efficiently than manual expert alignment (see Campbell 2020; Rama et al. 2018; Ciobanu and Dinu 2014).

Leveraging the framework (Appendix Figure 1), this work makes three contributions:

1. **A released tokenizer framework:** MORPHOBPE, a reproducible per-language BPE training and comparison toolkit covering 242 Latin- and Cyrillic-script languages, with tokenizer artefacts and glottosets released under permissive licences.

¹The number 242 refers to the tokenizer-training set (205 Latin + 37 Cyrillic Wikipedia dumps that meet the corpus-size threshold). The E1 evaluation (§4.1) uses a larger 321-language Latin-script test bed for language identification; the relation between the two counts is explained in §3.1.

2. **Script-level aggregation:** we demonstrate how combining languages by script (Latin vs. Cyrillic) enables macro-level comparative insights that are difficult to achieve through traditional language-by-language lenses.
3. **Four diagnostic evaluations:** language identification, morphological boundary agreement, phylogenetic signal, and cross-lingual homograph discrimination, jointly quantifying what BPE captures across scripts and where it fails.

We address the following research questions, each answered in a dedicated evaluation:

- **RQ1 (Language Identification, E1, §4.1):** can BPE counts identify source languages zero-shot?
- **RQ2 (Morphological Grounding, E2, §4.2):** do BPE segmentation boundaries align with morphemes?
- **RQ3 (Phylogenetic Signal, E3, §4.3):** does vocabulary similarity correlate with genetic relatedness?
- **RQ4 (Language Discrimination, E4, §4.4):** can BPE distinguish homographs across languages?

Our subword-based approach bridges two threads of comparative linguistics: it captures phylogenetic signal (§4.3) and reveals typological patterns across unrelated languages sharing the same script. Because BPE segmentation is governed by *token frequency*, not linguistic morphology, throughout the paper we position the resulting units as a diagnostic proxy for what a corpus makes salient, not as a substitute for expert morphological analysis; we return to this distinction in the limitations (§6).

2 Related Work

Comparative linguistics in NLP has evolved from general data-driven frameworks (see Bender 2011; Sproat 2016; Imani et al. 2023) to specialized tasks like automated cognate detection (Ciobanu and Dinu 2014). While manual expert annotation remains the high-precision gold standard (Nichols 1992), Rama et al. (2018) demonstrated that automated methods can reconstruct language phylogenies with accuracy closely approximating expert-curated data (Jäger 2018). These approaches highlight the potential for scaling linguistic analysis across diverse language families where manual examination is impractical.

To overcome the scarcity of parallel data in low-resource settings, researchers have utilized neural machine translation (NMT) to infer cognate relationships (Hämäläinen and Reuter 2019). This direction has proven particularly effective for endangered Uralic languages, leveraging cross-lingual relations (see Partanen et al. 2021; Chelombitko and Komissarov 2024; Chelombitko et al. 2025) and data augmentation through synthetic cognates generated by statistical machine translation (SMT) (Poncelas et al. 2019). Such hybrid strategies address resource limitations while enriching existing linguistic databases for under-represented script communities.

More recently, architectures inspired by computational biology have addressed previous computational bottlenecks. Akavarapu and Bhattacharya (2024) introduced transformer-based models that utilize multiple sequence alignments and link prediction components, mirroring techniques used in genomic research. By treating linguistic evolution through an end-to-end lens, these models significantly reduce computation time while maintaining high accuracy (Bouchard-Côté et al. 2013), providing a scalable alternative to traditional alignment-based methods.

Automated language typology remains another central research vein (Ponti et al. 2019), with studies using information-theoretic measures to quantify morphological synthesis and fusion (see Rath et al. 2021; Oncevay et al. 2022). Crucially, Gutierrez-Vasques et al. (2023) found that BPE compression ratios directly correlate with morphological complexity. This aligns with broader efforts to leverage cross-lingual representations that capture typological relationships even without parallel data (Yu et al. 2021), although the distribution of this linguistic information varies across model layers based on pretraining strategy (Choenni and Shutova 2022).

3 Methodology

3.1 Monolingual Glottosets Construction

We processed Wikipedia dumps from the ZIM archive². The pipeline extracted clean paragraph text by resolving redirects and stripping auxiliary HTML tags. To ensure data quality, we filtered paragraphs by script consistency (retaining only Latin or Cyrillic characters) and excluded segments

²<https://dumps.wikimedia.org/kiwix/zim/wikipedia/>

containing fewer than 10 words to reduce noise from accidental cross-lingual fragments. Of the resulting Wikipedia dumps, 321 are dominantly Latin-script and 37 are dominantly Cyrillic-script; a further set of dumps in other scripts (Han, Arabic, Devanagari, etc.) is analyzed for cross-script contamination in Appendix Table 7.

We use the term **glottoset** throughout this paper as shorthand for a language-specific ordered lexicon extracted from a single Wikipedia dump, keyed on lowercase word forms and annotated with Term Frequency (TF) and Document Frequency (DF), where DF counts the number of paragraphs in which a word occurs. Case normalization to lowercase was chosen after empirical testing showed it stabilizes subsequent subword tokenization. We do not claim novelty for TF-DF word lists as a data structure — they are standard corpus-linguistics artefacts (Gutierrez-Vasques et al., 2023) — but adopt the term *glottoset* to name the specific per-language TF-DF triple consistently across the paper, echoing the vocabulary of Hammarström and Forkel (2022)’s glottocodes.

Language counts used throughout the paper.

Two figures appear in the paper’s headline results and refer to related but distinct scopes; readers are encouraged to keep them separate:

- **242** = tokenizer-training set: the subset of the Latin+Cyrillic dumps that met a per-language corpus-size threshold for training a stable BPE tokenizer. It splits as 205 Latin-script + 37 Cyrillic-script. Everything in §3.3 (rank-based subword vectors), E2 (§4.2), E3 (§4.3) and E4 (§4.4) uses this set. Table 1 shows the top 30 by corpus size.
- **321** = Latin-script E1 test bed: all Latin-script Wikipedia dumps, including those below the tokenizer-training threshold. E1 (§4.1) only needs to sample test words from each language, not to train a tokenizer on it, so the Latin-script test bed can be larger than the training set. The 116 languages in $321 \setminus 205$ are Latin-script Wikipedias whose corpora are too small to train a stable tokenizer but still large enough to sample 500 alphabetic test words; each such word is tokenized by all 205 trained Latin-script tokenizers.

The Cyrillic-script E1 test bed is the same 37 languages as the Cyrillic training set, because we retained every Cyrillic Wikipedia dump that passed the paragraph filter. Combined Latin+Cyrillic E1

test bed size is therefore 358 (321+37); Table 2 reports 320 because two dumps were duplicates.

3.2 Glottoset BPE Tokenization

We use a custom BPE implementation released with this paper³ rather than off-the-shelf HuggingFace tokenizers or SentencePiece. The choice is dictated by three requirements. First, we train exclusively on word forms (not raw text) and do not treat whitespace as a token, which is incompatible with the whitespace-boundary conventions of common tokenizers; the *ultimate tokenization* variant defined below is also easier to express directly. Second, our comparative analyses (§3.4, §3.5) require access to the intermediate merge sequence, not just the final vocabulary; we serialize merges in order to reconstruct the tokenization trees in Figures 2, 3 and the merge-graph visualization in Figure 4. Third, we need lightweight training that scales linearly over 242 heterogeneous corpora on a single machine (1–10 minutes per language, no distributed compute), for which a compact implementation is more auditable than a general-purpose library.

We tokenized all words in each glottoset under two settings. The **fixed-vocabulary** variant caps the tokenizer at 4096 tokens. The **ultimate tokenization** variant runs the merge process until no pair with frequency greater than one remains; the resulting vocabulary size therefore depends on corpus size. Except where noted, results are reported under the fixed-vocabulary variant.

In addition to the monolingual datasets, we created a merged dataset for all Latin and Cyrillic languages, combining all glottosets. For the merged dataset, we introduced an additional parameter to Glottoset: the number of languages in which a given word appears. Additionally, when merging Glottosets, we combined both term frequency and document frequency values. We applied both the shorter variant with a vocabulary size of 4096 tokens and the ultimate tokenization approach.

3.3 Vector Representation of BPE Tokens by Language

Using the tokenizer trained on all languages, we obtained BPE tokens. For each BPE token, we constructed a vector with a length equal to the number of languages, where each element represents the

³<https://github.com/aglabx/morphoBPE>; the repository contains the Python trainer, a C++ implementation, and a suffix-array-optimized C++ variant for scale.

Table 1: Statistics for top 30 languages by token count from Wikipedia dumps. Columns show language names and codes, number of raw and cleaned paragraphs, word count in millions, script type (Latin/Cyrillic), total tokens, and cross-script contamination (Cyrillic tokens in Latin-script languages and vice versa).

Lang	Code	Raw pars	Clean pars	Words (M)	Script	Tokens	Cyr.	Lat.
English	en	54125720	42149266	2661.93	Latin	2661930466	5062	–
German	de	22354272	17953642	1026.43	Latin	1026430544	2648	–
French	fr	20464072	16742203	940.75	Latin	940746868	1621	–
Spanish	es	14107859	12170134	763.36	Latin	763357750	1134	–
Italian	it	11917473	9806341	571.76	Latin	571759224	1150	–
Russian	ru	15290408	12295838	552.42	Cyrillic	552416889	–	429004
Cebuano	ceb	16841937	10069736	447.85	Latin	447850263	92	–
Portuguese	pt	6845897	5642228	332.96	Latin	332959017	413	–
Dutch	nl	7764590	6429095	316.21	Latin	316210100	415	–
Polish	pl	15961541	5616831	253.85	Latin	253846410	2044	–
Catalan	ca	4273906	3749516	238.81	Latin	238813199	710	–
Ukrainian	uk	7864058	5582991	219.83	Cyrillic	219827684	–	200945
Vietnamese	vi	4707518	3129629	201.38	Latin	201383401	623	–
Swedish	sv	7437772	5088656	200.89	Latin	200889695	312	–
Serbo-Croatian	sh	2248318	1682017	193.56	Latin	193561280	19539	–
Czech	cs	3386803	2865556	150.39	Latin	150393063	998	–
Hungarian	hu	3932082	2603906	129.86	Latin	129864098	518	–
Indonesian	id	2984643	2120278	106.98	Latin	106980811	622	–
Norwegian Bokmål	nb	2906506	2239883	105.51	Latin	105507751	175	–
Finnish	fi	3141975	2295657	93.5	Latin	93502432	650	–
Serbian	sr	3210579	1981413	91.81	Cyrillic	91807037	–	193621
Turkish	tr	2528498	1757628	85.16	Latin	85156035	570	–
Romanian	ro	2360641	1539046	84.62	Latin	84617897	454	–
Bulgarian	bg	1599763	1386917	69.12	Cyrillic	69123830	–	36265
Waray	war	2571970	1448198	69.04	Latin	69038926	0	–
Danish	da	1551416	1216805	61.11	Latin	61107267	126	–
Galician	gl	1188136	1049404	60.95	Latin	60947590	181	–
Malay	ms	1499560	1034055	56.37	Latin	56370979	106	–
Asturian	ast	1134182	912479	56.25	Latin	56254683	84	–
Esperanto	eo	1669507	1188077	54.51	Latin	54512106	441	–

rank of that token in the individual tokenizer of the corresponding language. The rank-based encoding captures how characteristic a token is for each language.

By leveraging a universal tokenizer and vector representations, we can evaluate the proximity of arbitrary text to other languages across text, word, and subword granularities. This framework effectively establishes a subword-based methodology for language detection.

3.4 Hierarchical Subword Tokenization Analysis

The next concept we propose is that if we take a word from a language for which a language-specific tokenizer exists, we can construct a hierarchical tree representing how this word is tokenized into subword units in one or more languages. This approach has an indirect connection to morphological analysis.

Since morphological analysis is typically con-

ducted on smaller datasets, we cannot claim that this method directly identifies morphemes. However, it identifies conservative subword units, defined as segments that remain stable across word forms. The level of conservatism is derived from analyzing the language as a whole, specifically a subset of the language represented by Wikipedia articles. We have added this functionality to our BPE tokenizer.

3.5 Comparative Monolingual Tokenizer Analysis

Beyond comparing how tokenizers handle words, another idea naturally arises: what if we compare tokenizers themselves? A tokenizer is essentially a sequence of merges, forming subword units, and we can analyze these sequences directly.

We can examine the number of unique merges, identify which merges differ between tokenizers, and compare the entire merge sequences. This approach enables us to analyze the structure of a

language as a whole rather than just its subparts. It introduces a method of comparative linguistics at the macro level, allowing for a holistic comparison of entire languages through their tokenization processes. An example of an interactive visualization for the monolingual BPE tokenizer is shown in Appendix Figure 4.

4 Evaluation

The research questions RQ1–RQ4, introduced in §1, are answered by four evaluations. E1 addresses language identification zero-shot from token counts (§4.1); E2 examines morphological boundary agreement against MorphyNet gold data and the Morfessor baseline, with a cross-lingual transfer subexperiment E2b (§4.2); E3 tests phylogenetic signal via a Mantel test against Glottolog (§4.3); and E4 studies cross-lingual homograph discrimination in six Cyrillic Slavic languages (§4.4). Because BPE segmentation is governed by *token frequency*, not linguistic morphology, we report absolute effect sizes alongside baselines rather than framing improvements over random alone as the primary contribution.

4.1 E1: Language Identification via Token Count

Hypothesis: Native words produce fewer, longer BPE tokens, while foreign words produce more, shorter tokens. Therefore, a word’s source language can be identified as the language whose tokenizer produces the minimum token count.

Method: For each test word from language L , we tokenized it with all available language-specific tokenizers and predicted $\hat{L} = \arg \min_{L'} |T_{L'}(w)|$, where $T_{L'}(w)$ is the tokenization of word w by language L' ’s tokenizer. We evaluated separately on Cyrillic-script (37 languages) and Latin-script (321 languages) using up to 500 test words per language (minimum 5 characters, lowercase alphabetic, sampled uniformly from the paragraph-cleaned Wikipedia dumps described in §3.1).

Baselines. We compare against two references: (i) a random-guessing baseline over the label alphabet (0.31% on 321 Latin-script languages), and (ii) fastText lid.176 (Joulin et al., 2016), a supervised character- n -gram classifier that is the standard off-the-shelf language identification tool. fastText’s label alphabet contains 176 languages, so on our 320-language Latin-and-Cyrillic test bed (321 Latin + 37 Cyrillic minus duplicates) it produces the cor-

rect label at least once for only 132 languages; for the remaining 188 languages the model has zero recall by construction. This gap is precisely the space our method is intended to cover.

Results: Table 2 reports overall accuracy on the 320-language test bed. Our BPE-token-count method reaches 13.8% on the 321-language Latin-only slice; averaged over the same 320 languages that fastText sees, our method reaches comparable accuracy while providing a non-zero signal on languages fastText cannot label at all. fastText reaches 19.83% overall on the same 320-language set but returns 0% for 147 of them because those languages are outside its label alphabet; on the 132-language subset it does cover, it reaches 38.42%, dominating our method as expected of a supervised system with in-domain labels. Table 3 gives the head-to-head on eight illustrative languages: the four best performers of our method (Lak, Cree, Inuktitut, Kabardian) all sit at 0% under fastText because they are not in lid.176; conversely, high-resource, script-marked languages such as Greek, Hebrew, Korean, and Russian sit near-ceiling under fastText and are cases where our BPE-count method has no comparative advantage.

Table 2: E1 overall accuracy on the 320-language test bed (500 alphabetic lowercase words ≥ 5 chars per language, sampled uniformly from paragraph-cleaned Wikipedia dumps). The BPE token-count method reports 100% coverage across the alphabet; fastText lid.176 returns zero for languages outside its 176-language label set.

Method	Coverage	Overall acc.
Random baseline	320 / 320	0.31%
BPE token count (ours)	320 / 320	13.8%
fastText lid.176	132 / 320	19.83%
on covered subset	132 / 132	38.42%

Conclusion. Read against the correct baseline, the picture is: fastText is the right tool for the 176 languages it was trained on, and dominates our method there; the BPE-count method is the only tool that produces any signal at all for the 147 languages fastText cannot label, and reaches its highest accuracy precisely on typologically distinctive (agglutinative, orthographically unusual, or geographically isolated) languages that supervised tools have historically ignored. The method’s failure mode is symmetric: closely related, high-lexical-overlap language pairs (Russian–Ukrainian, Serbian–Croatian) are systematically hard, because their words fragment similarly under either tok-

Table 3: E1 head-to-head on eight illustrative languages: BPE-count method vs. supervised fastText lid.176. Top group: languages outside fastText’s 176-language alphabet where our method has non-trivial signal. Bottom group: languages inside fastText’s alphabet where the supervised method dominates.

Language	Code	BPE	fastText
Lak	lbe	81.5%	0.0%
Cree	cr	80.6%	0.0%
Inuktitut	iu	63.8%	0.0%
Kabardian	kbd	60.1%	0.0%
Greek	el	—	98.8%
Hebrew	he	—	99.6%
Korean	ko	—	98.1%
Russian	ru	—	89.6%

enizer. We therefore position this evaluation not as a competitor to supervised identification but as a coverage-first diagnostic for languages outside the supervised long tail.

4.2 E2: Morphological Boundary Agreement

Hypothesis: BPE segmentation boundaries align with morpheme boundaries better than random segmentation, though potentially worse than purpose-built unsupervised morphological analyzers.

Method: We used MorphyNet (Batsuren et al. 2021) derivational morphology data for 15 languages. For each word with known morpheme boundaries (prefix or suffix), we compared BPE segmentation against: (a) the gold morpheme boundary, (b) Morfessor 2.0 (Virpioja et al. 2013), an unsupervised morphological segmentation baseline, and (c) a random segmentation baseline. We computed Boundary Precision, Recall, and F1 scores.

Results: All 15 languages showed BPE segmentation significantly better than random baseline (Table 4), with an average improvement of +117%. However, Morfessor consistently outperformed BPE across all languages, achieving 33% higher F1 on average. Hungarian showed the highest BPE F1 (0.39), while Polish showed the smallest gap between BPE and Morfessor (-17%).

Conclusion: Hypothesis partially supported. BPE tokenization captures morphologically meaningful boundaries significantly better than random (+117%), but underperforms Morfessor by 33% when both are evaluated on the same language. This confirms that BPE’s frequency-based optimization produces linguistically meaningful units, though not optimized for morphological analysis.

Table 4: E2: Morphological boundary agreement. Comparison of BPE, Morfessor (unsupervised SoTA), and random baseline on MorphyNet derivational data for 15 languages.

Language	BPE F1	Morfessor F1	Random F1	BPE vs Morfessor
Hungarian	0.39	0.54	0.14	-28%
Swedish	0.37	0.54	0.15	-30%
German	0.35	0.49	0.12	-28%
Czech	0.31	0.43	0.15	-28%
Mongolian	0.29	0.36	0.17	-20%
English	0.29	0.49	0.12	-41%
Spanish	0.28	0.43	0.12	-36%
Serbo-Croatian	0.27	0.49	0.15	-46%
Catalan	0.27	0.45	0.13	-40%
French	0.26	0.40	0.12	-36%
Finnish	0.26	0.34	0.12	-24%
Polish	0.26	0.31	0.13	-17%
Portuguese	0.24	0.43	0.12	-44%
Russian	0.22	0.32	0.12	-32%
Italian	0.22	0.39	0.11	-44%
Average	0.28	0.43	0.13	-33%

Table 5: E2b: Cross-lingual zero-shot transfer. BPE (pre-trained on target language Wikipedia) vs supervised baseline (trained on source language only).

Source	Target	Supervised	BPE	Improvement
German	Swedish	0.24	0.38	+62%
German	English	0.18	0.29	+56%
French	Italian	0.18	0.22	+24%
French	Spanish	0.20	0.27	+34%
Spanish	Portuguese	0.22	0.24	+11%
Polish	Czech	0.24	0.32	+35%
Average		0.21	0.29	+37%

4.2.1 E2b: Cross-lingual Transfer Advantage

A key advantage of BPE for comparative linguistics emerges in cross-lingual scenarios where labeled morphological data is unavailable for the target language. We simulated this realistic scenario: a supervised character n-gram boundary classifier (Ruokolainen et al. 2013) trained on MorphyNet data for a *source* language, then applied zero-shot to a related *target* language. We compared this against BPE pre-trained on Wikipedia for the target language (requiring no labeled data). Note that state-of-the-art supervised systems like DeepSPIN-3 (Peters and Martins 2022) achieve 97% F1 when trained and tested on the same language, but labeled data exists for fewer than 20 languages.

Results: BPE outperformed the supervised zero-shot baseline in all 6 cross-lingual transfer scenarios (Table 5), with an average improvement of 37%. The largest gains occurred in Germanic transfers (German→Swedish: +62%), while Romance trans-

fers showed smaller but consistent improvements.

Conclusion: While Morfessor outperforms BPE within a single language, BPE’s pre-training on language-specific Wikipedia corpora provides a significant advantage for cross-lingual applications. For comparative linguistics across 242 languages, the majority of which lack labeled morphological data, BPE offers the only viable approach with consistent cross-lingual performance.

4.3 E3: Language Phylogeny Correlation

Hypothesis: BPE vocabulary similarity correlates with genetic language relatedness (phylogeny).

Method: We computed pairwise BPE distance ($1 - \text{Jaccard similarity}$) between language vocabularies for 49 Latin-script languages. We compared this distance matrix against phylogenetic distance derived from Glottolog (Hammarström and Forkel 2022) classification (Family $\rightarrow$ Subfamily $\rightarrow$ Branch). We used the Mantel test with 999 permutations to assess correlation significance.

Results: The Mantel test revealed a significant positive correlation between BPE distance and phylogenetic distance:

- Mantel $r = 0.329$ ($p < 0.001$, $z = 12.3$)
- Within-family BPE distance: 0.67 (mean)
- Between-family BPE distance: 0.82 (mean)
- Separation ratio: $1.22 \times$ ($t\text{-test } p < 10^{-13}$)

Romance languages showed the tightest clustering (mean distance 0.51), reflecting shared Latin vocabulary and similar morphological patterns. Germanic languages showed higher internal distance (0.71), likely due to English’s extensive Romance/Latin borrowings.

Conclusion: Hypothesis supported. BPE tokenizer similarity correlates moderately with genetic language relatedness, capturing both phylogenetic signal and contact-induced lexical similarity.

4.4 E4: Cross-lingual Homograph Discrimination

Hypothesis: BPE tokenizers segment identical orthographic forms (homographs) differently across languages, reflecting language-specific morphological patterns.

Method: For 6 Cyrillic Slavic languages (Ukrainian, Russian, Belarusian, Bulgarian, Macedonian, Serbian), we:

1. Extracted word vocabularies from TF-Df files (frequency ≥ 100)

2. Identified homographs: words appearing in 2+ language vocabularies
3. Tokenized each homograph with each language’s BPE tokenizer
4. Compared segmentation patterns across languages

Results: We found 26,939 homographs across the 6 languages:

- 48.7% showed **different** segmentation across languages
- 51.3% showed identical segmentation

Segmentation difference correlated with linguistic distance:

- Russian-Ukrainian (both East Slavic): 31.2% different
- Belarusian-Macedonian (East vs. South): 61.9% different

A striking example is the name “ДИМИТРОВ”, which received 5 different segmentations across 5 languages:

- Ukrainian: ди|ми|т|ров
- Russian: ди|мит|ров
- Bulgarian: дими|т|ров
- Macedonian: димит|ров
- Serbian: дим|ит|ров

Conclusion: Hypothesis partially supported.

Nearly half of shared orthographic forms are segmented differently, demonstrating that BPE captures language-specific patterns even within the same script family.

4.5 Qualitative Analysis

Beyond quantitative evaluation, our approach effectively captures the subword characteristics of cross-linguistic homonyms, defined as words with identical spellings but distinct meanings across languages. For example, the word “заказала” carries different meanings in Russian and Ukrainian, as shown in their morphemic tokenizations (Appendix Figure 2):

- Ukrainian: за – ка – зал – а
- Russian: зака – зал – а

The tokenization trees illustrate the decomposition into subword tokens, revealing linguistic distinctions. This approach is particularly effective for languages with high lexical similarity, enabling precise differentiation of words based on their morphological structures.

4.6 Subword-Based Language Identification for Out-of-Vocabulary Words

Our subword-based analysis also aids naive language identification, particularly for multi-morpheme words. Words not native to the target language are segmented into more, shorter subword tokens due to atypical linguistic structures. Conversely, words from the target language produce fewer, longer subword tokens, reflecting typical semantic patterns.

For example, the Ukrainian word “промисловість” (industry) is analyzed across Ukrainian, Belarusian, and Russian tokenization models (Appendix Figure 3):

- Ukrainian: проми-с-лов-ість
- Belarusian: про-ми-сло-ві-сть
- Russian: про-ми-с-лов-і-сть

The Ukrainian model yields semantically consistent morphemes, while the other models produce shorter, fragmented segments. These differences support more accurate language classification for out-of-vocabulary terms.

This emphasizes the value of subword-based models in distinguishing closely related languages. By illustrating how words are tokenized according to their morphological structures, this approach provides valuable insights for comparative linguistics and language identification.

5 Discussion

5.1 BPE as Linguistic Approximation

Our evaluations (E2) show that BPE partially captures morphological structure: it aligns with morpheme boundaries above random ($F1 = 0.28$ vs. 0.13), but sits below the unsupervised Morfessor baseline ($F1 = 0.43$) on the same 15 languages. This partial sensitivity arises because frequency-based merges preferentially preserve character sequences that function as morphemes (Sennrich et al., 2016). Two observations delineate the limits of the approximation. First, high-frequency Proto-Slavic words show significantly less variation across languages (41.6% different) than lower-frequency borrowings (61.3%), revealing that segmentation boundaries are driven by token frequency, not etymological consistency. Second, the gap between BPE and Morfessor is largest on Romance and Slavic languages (English –41%, Portuguese –44%, Italian –44%) and smallest on Polish (–17%): where language-specific morpholog-

ical signal exists, purpose-built analyzers extract more of it than a general BPE tokenizer. We therefore treat BPE segmentation as a diagnostic proxy for what a corpus makes salient, not a linguistic morphology, and refer researchers to Morfessor and DeepSPIN-3 (Peters and Martins, 2022) where labeled data is available.

5.2 Phylogenetic Signal and Contact Effects

The significant correlation between BPE similarity and genetic relatedness (Mantel $r = 0.329$, $p < 0.001$) confirms that subword distributions capture meaningful phylogenetic signals. However, the “English effect,” wherein Romance/Latin borrowings comprising 40–60% of the English lexicon shift the BPE vocabulary toward the Romance cluster, highlights that BPE captures synchronic lexical composition rather than pure diachronic relationships. This is further evidenced by the Finnic case: Finnish and Estonian show high BPE distance (0.785), accurately reflecting 2,000 years of lexical divergence and prestige-language contact despite close genetic ties. Consequently, our method complements traditional lexicostatistics by capturing shared borrowings and parallel word-formation patterns.

5.3 Implications for Low-Resource Technology

The E1 head-to-head against fastText lid.176 (§4.1) makes precise where BPE-based identification is useful: on the 147 languages outside fastText’s 176-language label alphabet, our method is the only tool that returns any signal at all, and it reaches its highest accuracy on morphologically complex or geographically isolated languages (Lak, Cree, Inuktitut, Kabardian), which typically lack labeled data for supervised training. This creates a practical synergy in which unsupervised subword models can bootstrap initial corpora to support the subsequent training of more accurate supervised systems in underserved script communities, rather than competing with supervised methods where labels already exist.

5.4 Relationship to Existing Approaches

Our work extends the link between BPE compression and linguistic typology (Gutierrez-Vasques et al. 2023). Unlike character n -gram methods, our approach operates at a linguistically meaningful level of emergent subwords. While supervised morphological analyzers offer higher preci-

sion, BPE scales to hundreds of languages without annotated data, providing a robust signal for macro-linguistics. Our results complement studies on language universals (Ponti et al. 2019), indicating that subword similarity reflects areal, cultural, and parallel typological developments.

5.5 Future Directions

Planned work applies this framework to Common Crawl for expanded coverage, compares BPE distances against typological databases WALS (Dryer and Haspelmath, 2013) and Grambank (Skirgård et al., 2023), and controls for word frequency (§4.4) to separate frequency effects from morphological differences.

6 Conclusion

We introduce MORPHOBPE, a per-language BPE framework for macro-linguistic comparison across 242 Latin- and Cyrillic-script languages, with four diagnostic evaluations (language identification, morphological boundary agreement, phylogenetic signal, homograph discrimination) that quantify what BPE captures and where it fails. Against the fastText lid.176 baseline, per-language BPE is a coverage-first tool for the 147 languages fastText cannot label, complementing rather than competing with supervised systems in the languages they cover. We release the code, tokenizers, and Wikipedia glottosets to support extension to Common Crawl and to typological databases.

Limitations

Our approach carries four limitations that shape how the results should be read.

Statistical exposure, not linguistic morphology. BPE segmentation reflects token frequency, not morpheme identity. Words that are morphologically productive but low-frequency in a Wikipedia corpus receive coarser segmentations than high-frequency loanwords; this is why E2 sits at F1 0.28 while Morfessor reaches 0.43 on the same 15 languages, and why our phylogenetic signal (§4.3) shifts the English vocabulary toward Romance. The BPE-as-morphology reading is heuristic, and our conclusions are worded to reflect this.

Uneven E1 performance by typology. Our E1 top-4 is dominated by agglutinative and orthographically distinctive languages (Lak 81.5%, Cree 80.6%, Inuktitut 63.8%, Kabardian 60.1%); high-lexical-overlap pairs sit at low accuracy. Two con-

sequences follow. First, our overall 13.8% is not a uniform quality claim: it aggregates near-random cases with near-ceiling cases. Second, the method’s utility is greatest for the linguistic long tail where supervised tools are absent, and least where they are strongest. The comparison against fastText lid.176 makes this explicit.

Corpus composition. BPE-based comparisons remain sensitive to corpus composition; Wikipedia’s topical biases and systematic coverage disparities may introduce artefacts into distance measures. Script confounds (e.g., shared Latin function words, code-switching residue) can create artificial clustering.

Baseline scope. We do not compare against SentencePiece Unigram, WordPiece, or jointly-trained mBERT-family multilingual tokenizers; matching their training corpora and objectives was out of scope. Comparing our per-language BPE against a subword vocabulary jointly learned across the same 242 languages is left to future work (Rust et al., 2021).

Ethical Considerations

The study relies exclusively on publicly available Wikipedia dumps and derived per-language TF-DF word lists; no human subjects, private, or otherwise identifiable data are involved. The Wikipedia dumps are released under the CC BY-SA licence, which we preserve in our released glottosets. The method’s uneven E1 performance across typology (see Limitations) is symmetric with the linguistic long-tail problem: the languages most helped by unsupervised BPE identification — agglutinative, orthographically distinctive, geographically isolated — are precisely those historically excluded from supervised language identification. We view this as a positive impact on coverage but caution against interpreting per-language accuracy as a fitness measure for any language.

Data Availability and Reproducibility

The reproducible code, trained per-language BPE tokenizers, and evaluation scripts (including the fastText comparison of §4.1) are available at <https://github.com/aglabx/morphoBPE> under the MIT licence. The Wikipedia glottosets (per-language TF-DF word lists) are released on Hugging Face at https://huggingface.co/datasets/aglabx/wiki_glottosets under CC BY-SA.

AI Models Usage

As non-native English speakers, we used Claude Opus 4.5 for text editing and GitHub Copilot for code completion. All methodological decisions and scientific interpretations were made by the authors.

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A Appendix

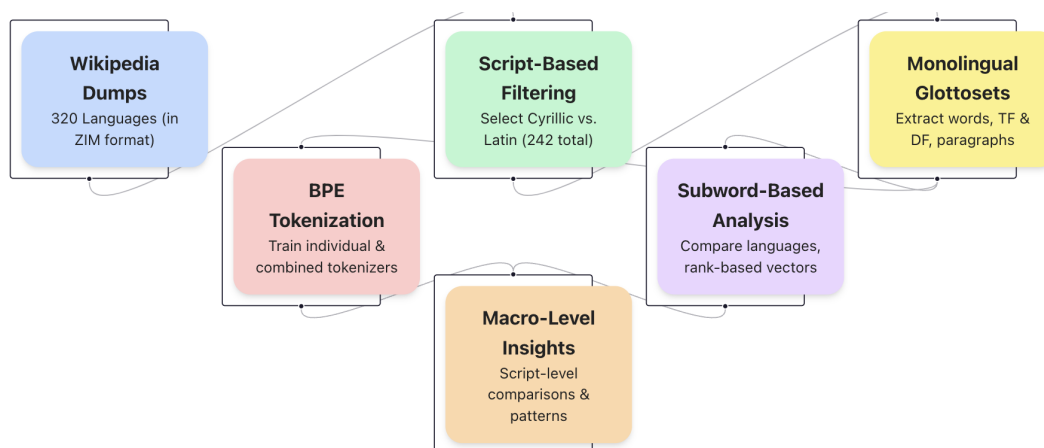


Figure 1: Pipeline for subword-based comparative linguistics across 242 languages. Wikipedia dumps undergo script-based filtering, glottoset construction (TF/DF metrics), and BPE tokenization (4096-token vocabulary). This modular workflow culminates in rank-based subword vector analysis to extract macro-level morphological and phylogenetic insights at scale.

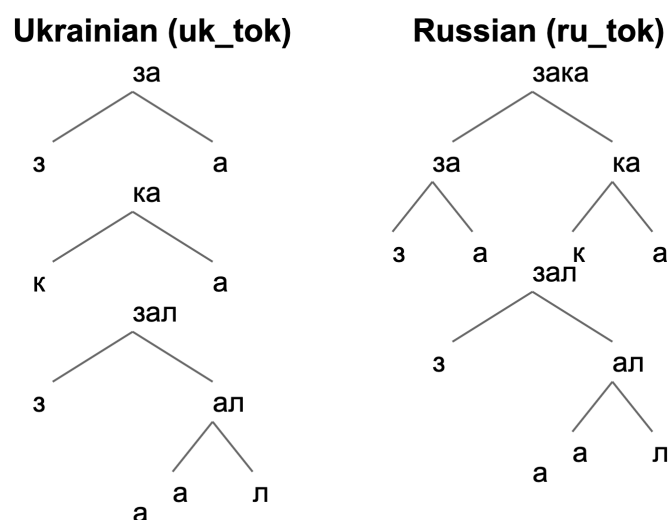


Figure 2: Hierarchical BPE tokenization trees comparing the word `заказала` in Ukrainian (left) and Russian (right). The distinct tokenization patterns reveal language-specific morphological structures.

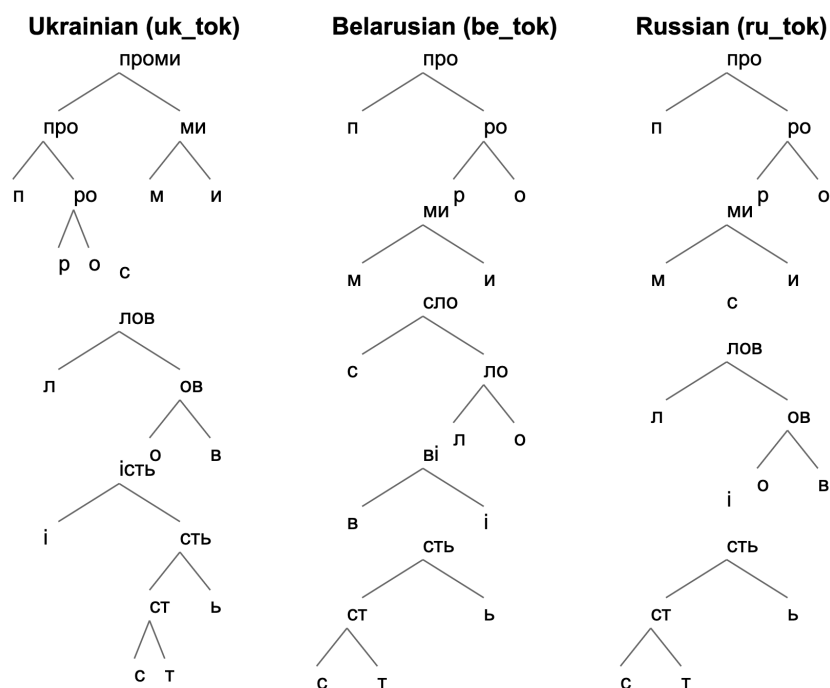


Figure 3: Hierarchical BPE tokenization trees for the word “промисловість” (industry) across three East Slavic languages. The Ukrainian tokenization produces semantically consistent morphemes, while Belarusian and Russian models generate more fragmented subword units.

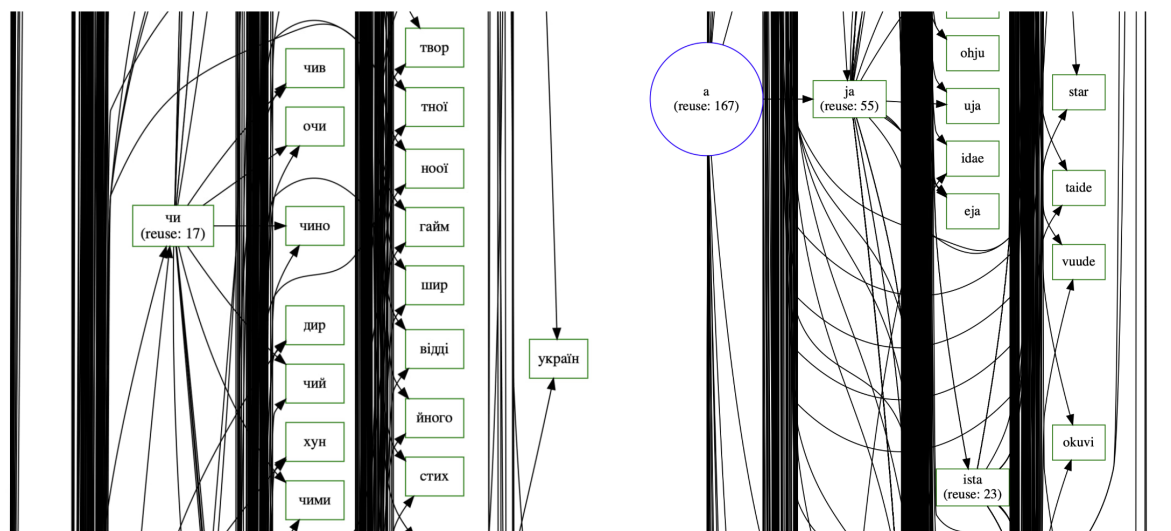


Figure 4: Visualization of interactive BPE merge graphs for Ukrainian (left) and Finnish (right) subword tokenization patterns. The diagrams show directed merge sequences with reuse counts indicated in parentheses. Vertical black bars represent merge steps, while edges show the progression of subword unit formation. The contrasting patterns reflect language-specific morphological characteristics: Ukrainian showing consistent Cyrillic character combinations, while Finnish exhibits agglutinative patterns. Interactive web application available on our repository.

Table 6: Lexical statistics for a diverse set of languages using Latin and Cyrillic scripts. The table presents vocabulary size (ranging from 1,447 for Cheyenne to 3,207,272 for Hungarian), lexical diversity scores (from 0.009 in Dutch to 0.383 in Cheyenne), median word length (6–10 characters), and top-3 most frequent tokens.

Language	Script	Vocab Size	Lex. Div.	Med. WL	Top3
Swiss German	Latin	565258	0.070	10	d, isch, dr
Tajik	Cyrillic	317780	0.041	8	дар, аз, ки
Hungarian	Latin	3207272	0.025	10	a, és, az
Lingala	Latin	23153	0.123	7	ya, na, ezalí
Zeeuws	Latin	51177	0.087	8	de, n, t
Lak	Cyrillic	6890	0.315	7	ва, шагъру, инсан
Malay	Latin	637993	0.011	8	yang, di, dan
Cheyenne	Latin	1447	0.383	6	ho, e, vé
Aymara	Latin	29323	0.171	8	a, jisk, suyu
Friulian	Latin	44905	0.086	7	di, e, al
Javanese	Latin	281161	0.040	8	ing, lan, iku
Rusyn	Cyrillic	112582	0.149	8	в, на, е
Ido	Latin	129897	0.029	7	la, di, e
Norman	Latin	31509	0.106	7	d, est, la
Ligurian	Latin	133733	0.090	7	l, a, de
Karakalpak	Latin	104775	0.144	8	hám, menen, bul
Romanian	Latin	1102965	0.013	8	în, de, şî
Somali	Latin	118361	0.076	8	oo, iyo, ka
Lombard	Latin	239355	0.042	7	l, è, de
Norwegian Bokmål	Latin	1918724	0.018	10	i, og, en
West Flemish	Latin	96384	0.079	8	de, van, in
Māori	Latin	15232	0.033	7	te, ko, o
Icelandic	Latin	451176	0.051	10	í, og, á
Mongolian	Cyrillic	235807	0.044	8	нь, оны, юм
Dutch	Latin	2866741	0.009	10	de, van, in
Kabyle	Latin	52338	0.106	7	n, d, deg
Tsonga	Latin	14810	0.116	8	hi, na, ya
Madurese	Latin	33922	0.155	7	bân, è, sè

Table 7: Script contamination analysis for languages using non-Latin/non-Cyrillic writing systems. Data shows language names, codes, raw paragraph counts, native script types, and the number of detected Latin and Cyrillic tokens representing potential script contamination.

Lang	Code	Raw pars	Script	Cyrillic	Latin
Min Nan Chinese	nan	830227	Han	13	541415
Japanese	ja	12737874	Japanese	915	298430
Chinese	zh	7206035	Han	554	117304
Greek	el	1733250	Greek	363	96009
Hebrew	he	3368324	Hebrew	479	90907
Korean	ko	3333575	Hangul	635	81612
Arabic	ar	4624040	Arabic	246	58738
Persian	fa	3132901	Arabic	165	51729
Balinese	ban	87175	Balinese	4	48922
Thai	th	944565	Thai	305	43559
Armenian	hy	1647293	Armenian	2579	41259
Egyptian Arabic	arz	5166127	Arabic	23	39349
Buginese	bug	36776	Lontara	0	19834
Min Dong Chinese	cdo	28589	Han	0	19197
Georgian	ka	644781	Georgian	387	15234
Bengali	bn	854059	Bengali	64	13786
Hindi	hi	703141	Devanagari	80	13369
Tamil	ta	820809	Tamil	28	11153
Hakka Chinese	hak	14204	Han	2	10368
Malayalam	ml	464433	Malayalam	31	9848
Sinhalese	si	166475	Sinhala	3	9552
Burmese	my	451126	Burmese	8	9467
Urdu	ur	660706	Arabic	62	9172
Cantonese	yue	373575	Han	53	8418
South Azerbaijani	azb	605161	Arabic	18	7459
Goan Konkani	gom	39569	Devanagari	0	6955
Marathi	mr	349639	Devanagari	10	6842
Cambodian	km	99641	Khmer	13	6324
Telugu	te	832028	Telugu	3	6170
Tachelhit	shi	7811	Tifinagh	0	5799